

Dear Ross,

I agree with your analysis of the CrowdStrike incident. It demonstrated the consequences of insufficient consideration of human capital and organisational oversight within the software and digital infrastructure sector. From an Industry 5.0 perspective, three dimensions of human capital development are particularly relevant in the context of information systems failures: trust in AI systems, as a prerequisite for effective human–machine collaboration; algorithmic explainability, which enables managers and developers to validate system outcomes; and risk management capabilities in automated production environments when anomalies or ethical concerns emerge (Pant *et al.*, 2025; Xu *et al.*, 2025). Greater integration of these principles may have helped either prevent the incident or reduce its negative impacts on both Microsoft and affected customers.

You are also correct in highlighting the importance of combining technology and human capability to ensure that data analytics generates organizational value, as well as the need for reliable data conditions for the successful adoption of machine learning in industrial environments. Although the CrowdStrike incident was not caused directly by an automated data-driven decision, but rather by systemic failures in both manual and automated testing and validation processes (Meyers, 2024), it nevertheless illustrated the broader risks associated with highly automated cybersecurity ecosystems, where CrowdStrike is a renowned player (Alspach, 2026). As Industry 4.0 security systems become increasingly interconnected, these interdependencies create new vulnerabilities, whereby failures within one system can rapidly propagate across others, potentially leading to large-scale cascading disruptions (Sturaro *et al.*, 2020).

Apart from the measures already discussed, another key preventive approach would have been the implementation of a safer privilege architecture. CrowdStrike Falcon Sensor operates with kernel-level access within Microsoft Windows systems. While this level of access enables real-time threat detection, it also increases systemic risk because faulty updates can affect critical operating system processes (Mugu *et al.*, 2024). Through applying the principle of least privilege by limiting software access to essential functions and isolating update mechanisms from kernel-critical operations (Saltzer and Schroeder, 1975), Microsoft engineers could have reduced the scale of the disruption.

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